# JAVASCRIPT BASICS & DESIGN EXAM

JavaScript Paper Exam-1

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**Course:** JavaScript MCQ (Chapters 1 & 2)

**Date:** ____________________________

> **Instructions:** This exam paper consists of 25 multiple-choice questions covering JavaScript syntax, design conventions, variables, datatypes, and scripting environments. Read each question carefully and select the single best option. Write your answers on a separate sheet or circle them directly on this paper. An Answer Key is appended on the final page of this booklet.

### Q1. Identify the outcome of this immediate script when loaded in a browser:

```
<script language="JavaScript">
document.write("Hello designers and developers.");
</script>
```

**A)** It waits for a button click event before displaying the text on the page.

**B)** It executes immediately as soon as the browser parser passes over it.

**C)** It fails to run because JavaScript cannot be written directly inside the <body>.

**D)** It automatically redirects the user to a new HTML page.

### Q2. What error in the following script will cause the browser's interpreter to ignore or fail to execute the JavaScript code?

```
<script language="JavaScript">
Alert("Pay attention to your cases!");
</script>
```

**A)** The string parameter must be enclosed in single quotes.

**B)** Semicolons are strictly mandatory at the end of every line.

**C)** The capitalization of "Alert" invalidates the function because JavaScript is case-sensitive.

**D)** The <script> tag is missing its version number attribute.

### Q3. According to the text, what is a fundamental difference in how HTML and JavaScript handle letter casing?

**A)** HTML is case-sensitive, while JavaScript is case-insensitive.

**B)** HTML is case-insensitive, while JavaScript is case-sensitive.

**C)** Both HTML and JavaScript are completely case-sensitive.

**D)** Neither HTML nor JavaScript pays attention to letter casing.

### Q4. Which of the following represents a legitimate variable name under JavaScript naming rules?

**A)** var Bill Sanders = "JavaScript guy";

**B)** var break = "coffee";

**C)** var Bill_Sanders = "JavaScript guy";

**D)** var 1stPlace = "Winner";

### Q5. Which keyword must be used to signal the initial creation of a variable in a script to prevent local scope issues?

**A)** dim

**B)** var

**C)** let

**D)** new

### Q6. What is meant by describing JavaScript as a "weakly typed" or "untyped" language?

**A)** It lacks the strength to handle mathematical operations like multiplication.

**B)** It requires explicit conversion steps for every single data type calculation.

**C)** It automatically figures out the type of data and makes the necessary adjustments without explicit type declarations.

**D)** It only allows variable names that begin with lowercase letters.

### Q7. What is the final output of the following weakly typed script when run?

```
<script language="JavaScript">
var apples = 1.43;
var oranges = 2.33;
var subtotal = apples + oranges;
var message = "Your total is $";
var deliver = message + subtotal;
document.write(deliver);
</script>
```

**A)** 3.76

**B)** Your total is $3.76

**C)** It displays nothing because you cannot add a string to a number.

**D)** Your total is $apples + oranges

### Q8. What is the function of the double forward slashes (//) in the following code block?

```
//Include a variable to add taxes
tax = .06;
```

**A)** They perform integer division on the variable tax.

**B)** They mark the line as a comment, which is ignored by the browser's parser.

**C)** They act as an alternative syntax for the closing script tag.

**D)** They set the scope of tax as a local variable.

### Q9. Why is it conventionally recommended to place JavaScript function definitions in the section of an HTML document?

**A)** The <body> section cannot execute immediate scripts.

**B)** All code in the head section is loaded first, ensuring functions are fully loaded before a user can fire them.

**C)** The head section automatically translates JavaScript into a compiled language.

**D)** Placing code in the head hides the script from being viewed via "view source".

### Q10. What is the primary role of the web browser in executing an "interpreted language" like JavaScript?

**A)** It compiles the script into an executable file before loading the page.

**B)** It acts as a translator, parsing (interpreting) the code line by line and instructing the computer to execute it.

**C)** It sends requests to a remote database server to compute every line of code.

**D)** It restricts execution so that only case-insensitive syntax is permitted.

### Q11. Examine this deferred script. When does the background color of the document change?

```
<input type="button" value="#d0d0a9" onClick="document.bgColor='d0d0a9'">
```

**A)** Immediately as soon as the parser loads this line of HTML.

**B)** When the page finishes loading entirely.

**C)** Only when the user clicks the button, firing the event handler.

**D)** When the user moves the mouse cursor completely off the page.

### Q12. How many arguments does the built-in prompt() function typically take, and what do they represent?

**A)** One argument, representing only the title of the input box.

**B)** Two arguments: the prompt message and the optional placeholder text.

**C)** Three arguments: the prompt message, a placeholder, and a background color.

**D)** It takes no arguments because it is a primitive data type.

### Q13. Which of the following is a reserved word in JavaScript that cannot be used as an identifier for variables or functions?

**A)** apple

**B)** customers

**C)** total

**D)** break

### Q14. What is the official standardization name for JavaScript revisions (such as versions 1.2, 1.3, and 1.5) that browser interpreters adhere to?

**A)** HTML-300

**B)** ECMA-262

**C)** W3C-CSS

**D)** Java SDK

### Q15. According to the author, why should designers learn to write custom JavaScript instead of solely relying on visual design tools to generate script?

**A)** Generated code cannot support mouse event handlers like onMouseOver.

**B)** Generated code often causes "code bloat," which increases file size and makes pages load slower.

**C)** Visual tools can only generate case-insensitive scripts that crash browsers.

**D)** JavaScript generated by design tools only works on the UNIX operating system.

### Q16. What is the behavior of semicolons in a JavaScript program?

**A)** They are strictly mandatory; omitting one always halts program execution.

**B)** They are only allowed within curly braces in a function.

**C)** They are optional because JavaScript places "invisible" semicolons at line breaks, but using them is highly recommended for clear debugging.

**D)** They are used exclusively to denote comments.

### Q17. To prevent very old browsers (which do not understand JavaScript) from displaying script code as plain text on a web page, which characters should be used to wrap the script inside the script tag?

**A)** /* and */

**B)** // and //

**C)** <!-- and //-->

**D)** <?php and ?>

### Q18. If a local variable inside a function is given the exact same name as a global variable, how does the function treat the variable name?

**A)** It throws a duplication error and crashes.

**B)** The local variable takes precedence over the global variable within that function.

**C)** It ignores the local variable and uses the global variable's value.

**D)** It automatically converts both variables to string types.

### Q19. What will the alert box display when this script is run?

```
<script language="JavaScript">
var localGlobal = "I'm global now!";
function showMe() {
var localGlobal = "I'm now local";
alert(localGlobal);
}
showMe();
</script>
```

**A)** I'm global now!

**B)** I'm now local

**C)** undefined

**D)** Nothing; an error is thrown

### Q20. What is the risk of omitting the keyword var when declaring a local variable inside a function?

**A)** The variable is restricted to case-insensitive evaluations.

**B)** The program might interpret it as a change to an existing global variable's value rather than creating a local variable.

**C)** The variable will return a "null" literal error.

**D)** Semicolons will become mandatory for that specific function.

### Q21. What happens if you mismatch the case configuration of a variable when referencing it later in a math calculation, as shown below?

```
<script language="JavaScript">
var Tax = .05;
var Total = 10 + (10 * tax);
</script>
```

**A)** JavaScript ignores the casing difference and calculates .05 correctly.

**B)** It will fail or produce incorrect results because Tax and tax are treated as completely different identifiers.

**C)** The variable tax automatically defaults to the primitive value 0.

**D)** The double forward slashes will automatically comment out the line.

### Q22. According to Chapter 3, what is the definition of a "literal" in JavaScript?

**A)** An object-oriented property that cannot be changed.

**B)** Raw data that represents literally what it is, making up the root of data types (such as numbers, strings, and Boolean values).

**C)** A specialized variable containing the URL path of the document.

**D)** The invisible semicolons that terminate a script line.

### Q23. Unlike alert(), which formats line breaks using the escape sequence \n, the document.write() method breaks lines on a page using which formatting technique?

**A)** The \n new line sequence.

**B)** Standard HTML formatting tags, such as <br>.

**C)** Invisible semicolons.

**D)** Pre-compiling the code through a double forward slash.

**Q24. Fill in the blanks to make this immediate script tag container correct and workable:**

```
<html>
<body>
<_______ language="JavaScript">
document.write("Hello designers and developers.");
</_______>
</body>
</html>
```

**A)** head

**B)** body

**C)** script

**D)** var

**Q25. Rearrange the following blocks of code to construct a workable script that outputs "Welcome, Willie" on the screen when loaded:**
**• Block 1: document.write("Welcome, " + yourname);**
**• Block 2: </script>**
**• Block 3: var yourname = "Willie";**
**• Block 4: <script language="JavaScript">**

**A)** Block 1 --> Block 3 --> Block 4 --> Block 2

**B)** Block 4 --> Block 3 --> Block 1 --> Block 2

**C)** Block 4 --> Block 1 --> Block 3 --> Block 2

**D)** Block 3 --> Block 4 --> Block 1 --> Block 2

# EXAM ANSWER KEY

The following table displays the correct option and a brief explanation/chapter context for each of the 25 questions in this exam. These answers correspond to the teachings of *JavaScript Design* (William B. Sanders, 2001) and general web standards documented on W3Schools.

| Q# | Key | Explanation / Chapter Context |
|---|---|---|
| 1 | **B** | Immediate scripts execute as soon as the browser parses them, before the rest of the page loads. |
| 2 | **C** | JavaScript is case-sensitive; function 'Alert()' is unrecognized (built-in function is lowercase 'alert'). |
| 3 | **B** | HTML is case-insensitive, but JavaScript is strictly case-sensitive. |
| 4 | **C** | Legitimate variable names cannot contain spaces, must not start with a digit, and cannot be reserved keywords. 'Bill_Sanders' is legal. |
| 5 | **B** | The keyword 'var' initializes a variable; omitting it inside a function can cause accidental global variable overrides. |
| 6 | **C** | Weakly typed languages do not require explicit variable data type declarations and perform conversions automatically. |
| 7 | **B** | The code adds the strings and values: apples (1.43) + oranges (2.33) = subtotal (3.76). Concatenated with message gives 'Your total is $3.76'. |
| 8 | **B** | Double slashes (//) indicate a single-line comment, which is completely ignored by the parser. |
| 9 | **B** | Placing function definitions in the head section ensures they are completely loaded before the user can trigger them. |
| 10 | **B** | Interpreted languages are not pre-compiled; the browser interprets the script line-by-line during runtime. |
| 11 | **C** | The onClick event deferment means the code executes only when triggered by user interaction (clicking the button). |
| 12 | **B** | The prompt() function accepts two parameters: the prompt message and the optional default placeholder text. |
| 13 | **D** | Keywords like 'break' are reserved for control flow structures and cannot be used as variable identifiers. |
| 14 | **B** | ECMA-262 is the official international specification standard for JavaScript revisions. |
| 15 | **B** | WYSIWYG layout editors often inject excessive visual helper routines leading to slow, bloated scripts ('code bloat'). |
| 16 | **C** | Semicolons are optional in JS due to automatic semicolon insertion at line breaks, but are highly recommended. |
| 17 | **C** | Hiding scripts inside '<!--' and '//-->' prevents legacy browsers from displaying JavaScript code as plain page text. |
| 18 | **B** | Local variables inside a function scope shadow/override global variables of the same name within that function. |
| 19 | **B** | The shadowed 'localGlobal' inside showMe() takes precedence, alerting 'I'm now local'. |
| 20 | **B** | Omitting 'var' inside a function dynamically reassigns an existing global variable rather than creating a local scoped one. |

| Q# | Key | Explanation / Chapter Context |
|---|---|---|
| 21 | **B** | Case sensitivity means 'Tax' and 'tax' are separate variables; reference to undefined 'tax' causes NaN or runtime failure. |
| 22 | **B** | A literal is raw, unassigned data directly typed into code, representing exactly what it is (e.g. 500, 'text'). |
| 23 | **B** | document.write() writes directly to the HTML DOM, requiring standard HTML line break elements like '<br>' to format breaks. |
| 24 | **C** | The script container tags are '<script>' and '</script>'. |
| 25 | **B** | Correct ordering: Block 4 (<script>), Block 3 (var definition), Block 1 (document.write), Block 2 (</script>). |